Filtering with the DFT

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Key Concepts

1) Implementing a linear time-invariant filter using convolution produces one output sample for each new input sample.

2) Implementing a linear time-invariant filter using the DFT involves taking the DFT of the input, multiplying by the DFT of the filter impulse response, and taking an inverse DFT.

    a) The DFT approach requires a block of input data and produces a block of output data.

b) The DFT length, N, must be greater than or equal to the length of the input plus the length of the filter impulse response minus one.

c) The DFT approach can be faster than ordinary convolution because of the availability of computationally efficient algorithms for computing the DFT.

3) The DFT approach can be used when the input is not of finite duration by breaking the input up into sections, using the DFT to compute the convolutions for the individual sections, and combining the convolutions.

Video Screenshots

Filtering With the DFT

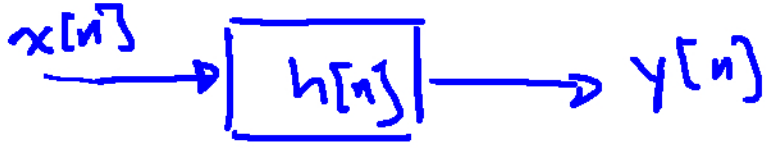
$$y[n] = x[n] * h[n] = \sum_{k=0}^{M_h-1} h[k]\,x[n-k]$$

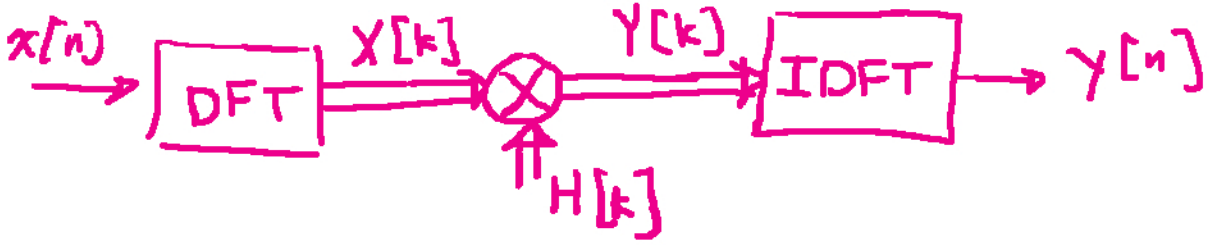
$$\xleftrightarrow{\text{DTFT}}$$

$$Y(e^{j\omega}) = X(e^{j\omega})\,H(e^{j\omega})$$

$$\downarrow \text{DFT}$$

$$Y[k] = X[k]\,H[k]$$

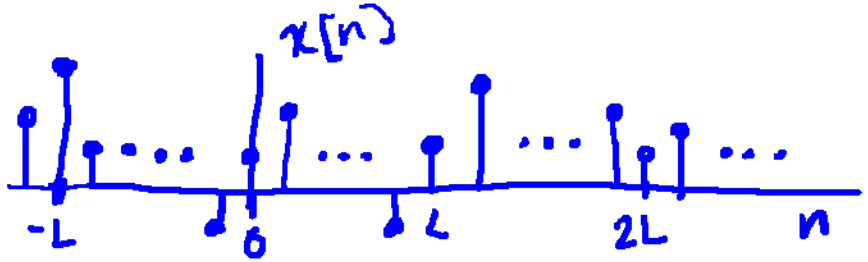




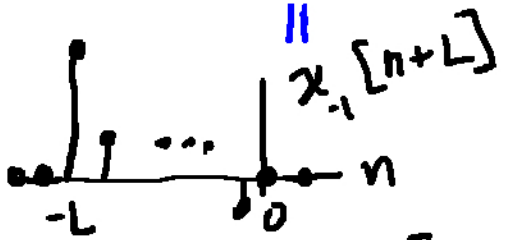
- Compute one output sample ($y[n]$) at a time

- Block of $x[n]$ $\longrightarrow$ Block of $y[n]$
- DFT length $N \geq M_h + M_x - 1$
- Can be "faster"

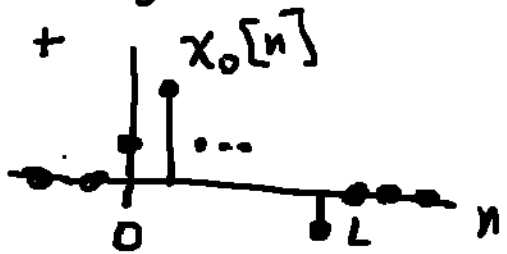
What if x[n] is not finite duration?



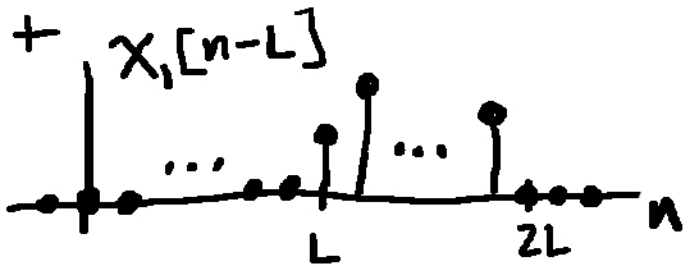
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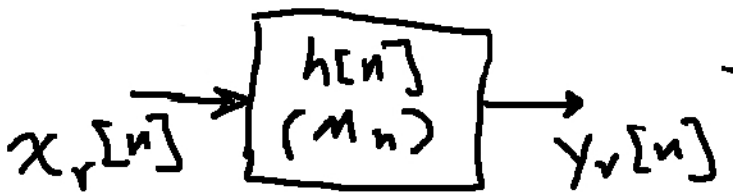
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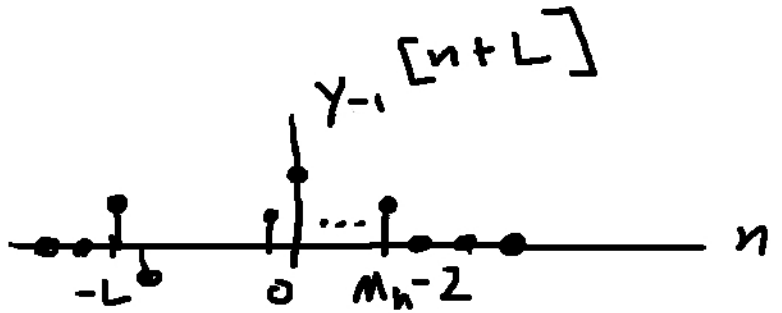
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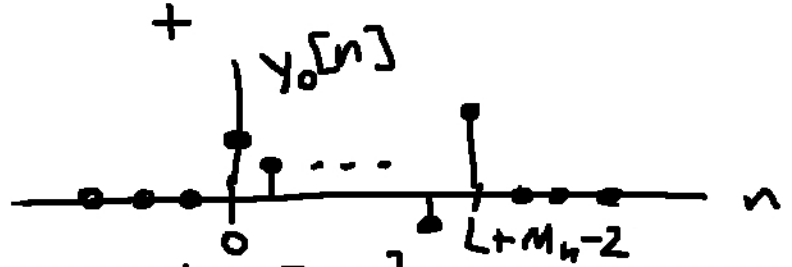
$$x[n] = \sum_{r=-\infty}^{\infty} x_r[n-rL]$$



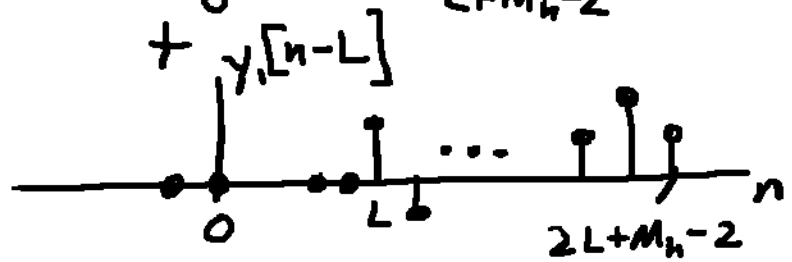
$$y[n] = x[n] * h[n]$$
$$= \sum_r x_r[n-rL] * h[n]$$
$$= \sum_r y_r[n-rL]$$



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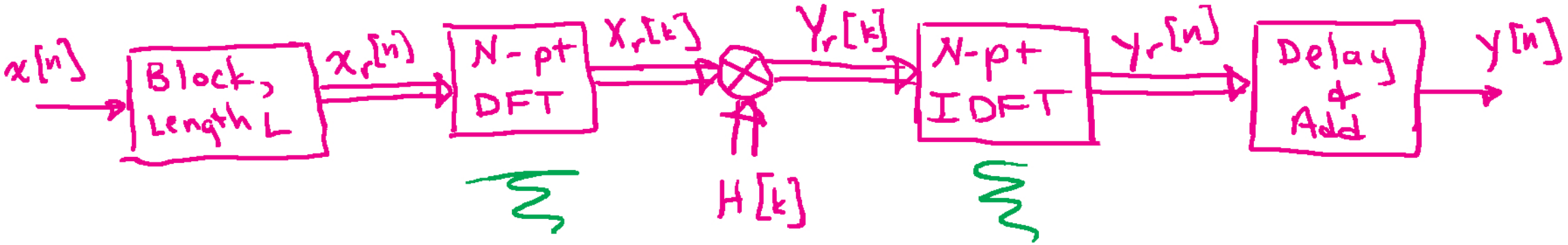
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Called: "Overlap and add" method

1) Calculate $y_r[n]$ using $N \geq L + M_h - 1$ point DFTs

2) Add up shifted $y_r[n]$: $y[n] = \sum_r y_r[n - rL]$



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